

Capstone hypothesis and baseline

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Documenting a network experiment

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset shift in neural models

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Similarity and normalisation

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Calculate the reference score for Capstone hypothesis and baseline. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Capstone hypothesis and baseline. Proposed explanation: “A hypothesis is a result”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Plan a new case independently

A new task predicts $\max(0, a \cdot x)$, with baseline $a = 1$. Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate baseline validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Write the hypothesis

State the prediction rule, baseline, candidate learning rate and update count, comparison data and metric. Write a testable hypothesis.

My reasoning and evidence

6. W2 Calculate the reference score

Compute baseline predictions, errors, squared errors and MSE on V1 and V2. State the denominator.

My reasoning and evidence

7. W3 Audit the plan

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

My reasoning and evidence

8. W4 Plan a new case independently

A new task predicts $\max(0, a \cdot x)$, with baseline $a = 1$. Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate baseline validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My reasoning and evidence

9. W5 Preserve the reserved evidence

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent evidence.

My reasoning and evidence

10. Project checkpoint and reflection

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final evidence pack: A new task predicts $\max(0, a \cdot x)$, with baseline $a = 1$. Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate baseline validation MSE and write a strict improvement criterion. Has the candidate succeeded yet? Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Capstone hypothesis and baseline

A testable image-pattern hypothesis. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Capstone hypothesis and baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write the prediction

B should have more correct labels than A on the same eight patterns.



2. Declare the baseline

A is a documented corner-only network, not a deliberately broken version.



3. Reserve evaluation

Keep the eight patterns out of model selection and tuning.

CHANGE ONE THING

Give only B the eight evaluation labels during training.

Explain the changed result using the matching lesson.

Name the intended change, the baseline and the measured outcome. Why must both versions avoid training on the eight reserved labels?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.